LIQUID-LIQUID EXTRACTION

Pre-Lab Plan
Unit Operations Lab 6
4/7/2026

Group #1, Tuesday 1PM section

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1. Experimental Objective (5pts)

The objective of this experiment is to investigate the extraction of acetic acid from mineral oil using water in a three-stage countercurrent mixer-settler system with variable speed agitators. The concentrations of acetic acid in both phases will be measured and plotted to construct a McCabe-Thiele diagram for evaluating the efficiency of the separation. Murphree efficiencies will also be calculated to assess the performance of each stage, in all three trials, as well as the overall efficiency of the system.

2. Experimental

For this experiment, liquid-liquid extraction will be studied in a two-phase system consisting of mineral oil containing acetic acid as the feed and water as the countercurrent extracting solvent. The two phases will be contacted and mixed thoroughly in a three-stage mixer settler system using variable speed agitators. The system will be allowed to reach steady state so that equilibrium mass transfer of acetic acid between the two phases can occur in each stage.

Samples from both the oil and water phases will be collected from each stage and analyzed by titration with 0.05 N NaOH to determine acetic acid concentrations. The experiment will be repeated at three different agitator speeds. The resulting concentration data will be used to construct McCabe Thiele diagrams to estimate the theoretical number of stages required for the system. These data will also be used to calculate Murphree efficiencies for each stage as well as the overall efficiency.

2.1 Apparatus (5pts)
The system is a Denver Mixer Settler Unit with a mineral oil/acetic acid mixture. The apparatus contains multiple impellers in different stages to conduct extraction from separation. Below is a labeled image of the apparatus with components labeled. A table below is provided to illustrate the purpose of each component within the apparatus.

Image 1: Image of Apparatus

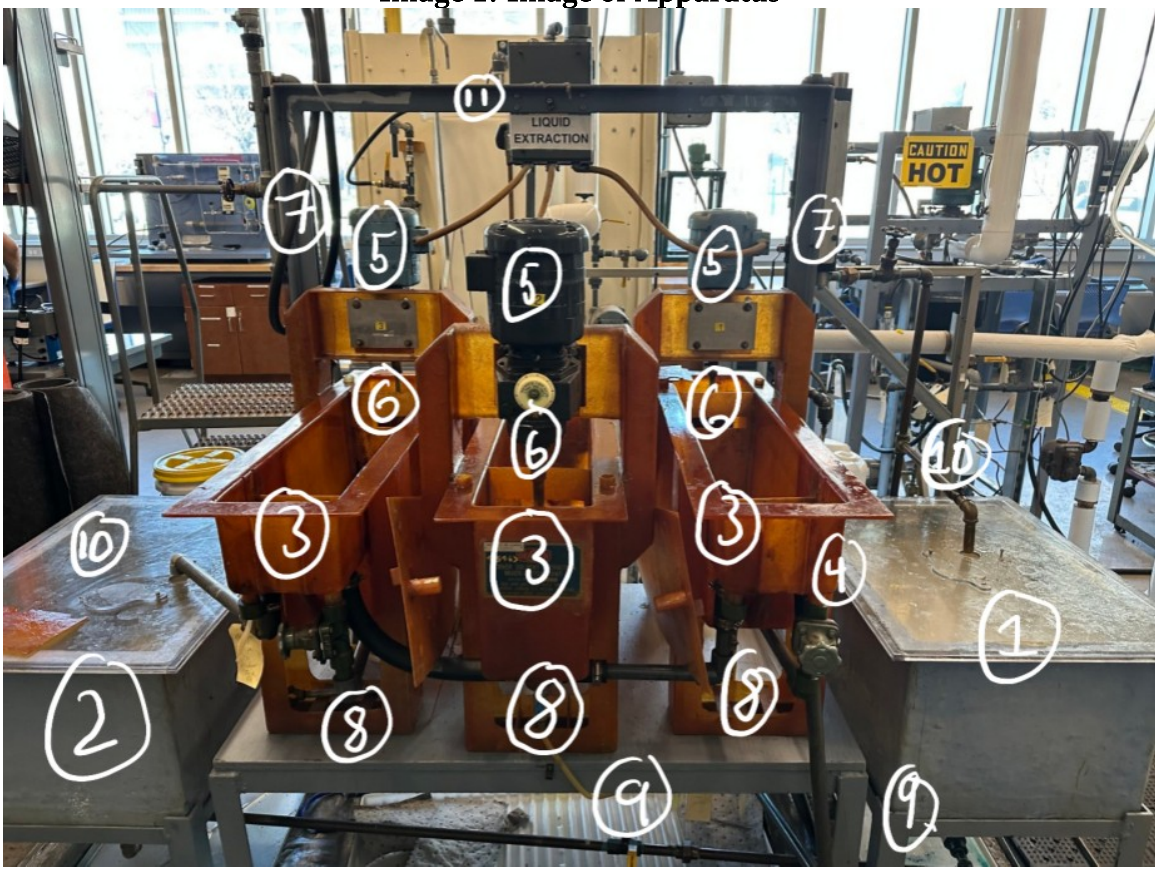

Table 1: Apparatus List

Component Number:	Component Name:	Description of Component:
1	Reservoir Feed Tank	Contains 25 gallons of mineral oil and acetic acid mixture to run the extraction experiment.
2	Product Collection Tank	The tank collects the processed mineral oil and acetic acid mixture product after the product runs through the experiment.
3	Denver Mixer Settler Unit	The main unit where the experiment is conducted. The unit has three extraction stages that each contain a mixer/settler unit within.
4	Centrifugal Pump	Used to mix the feeds from the tank into the

		extraction units of the denver mixer settler.
5	Motors	Three motors are used to control the speed of the impellers for the mixing application of the denver mixer settler.
6	Impellers	Three impellers are placed within the denver mixer settler, that are controlled with a in system circuit breaker to mix the mineral oil and acetic acid mixture. The speed of the impeller rotations is measured with the tachometer.
7	Rotameters	Two rotameters are on both sides of the denver mixer settler to control the flow rates of water and mineral oil/ acetic acid mixture.
8	Reflux Valves	Reflux valves control the reflux flow. The reflux valves are found under each extraction unit of the denver mixer settler. (Keep valves **fully closed** during the experiment)
9	Recycle Valves	Yellow valve that is found below the feed tank.
10	Lids	Placed above lids while conducting experiments to prevent spills. Only remove mixtures for titration analysis.
11	Solvent Cricut	Inlet used to supply water into the system from the top of the system.

2.2 Materials and Supplies (5pts)

Table 2: Materials and Supplies

Material/Supply	**Description**	**Material or Supply**
Mineral Oil (White Oil)	The liquid solution containing the 1.0 wt% acetic acid.	Material
Water	The solvent that extracts acetic acid from the white oil in the three-staged mixer-settler system.	Material
Acetic Acid	The solute added to white oil.	Materials
NaOH Solution	The titrant used to measure acetic acid concentration in mineral oil and water samples.	Material
Phenolphthalein Indicator	The indicator used to determine the endpoint when titrating.	Supply
Flasks/Beakers	Used to hold the raffinate and extract samples from each stage until the titration stage.	Supply
Graduated Cylinder	This measures the flow rates of the	Supply

	raffinate and extract phases.	
Magnetic Stirrer/Stir Bar	This makes the non-aqueous acetic acid and oil mixture easier to titrate with the aqueous NaOH titrant.	Supply
Burette	This delivers the 0.05 N NaOH used as the titrant.	Supply
Parafilm	Seals the samples waiting to be titrated and prevents acetic acid from evaporating out of solution.	Supply
Lab Tape/Marker	Used to label glassware and other utensils/equipment used in the lab.	Supply

2.3 Experimental Plan (10pts)

PROCEDURE:

1. Start by ensuring that there's ~25 gallons of feed in the tank and that its composition is ~1.0 wt% acetic acid by performing an initial titration with 0.05 N NaOH (equivalent to 0.05 M NaOH). Based on the results from the initial titration, correct the feed composition or continue with the next step.
    a. **NOTE:** Be sure to thoroughly mix the feed by using the centrifugal pump before taking a sample to titrate; turn the pump off before taking a sample. Also, always keep the cover on the feed tank, when possible, to prevent acetic acid from evaporating.
2. *IMPORTANT*: Keep a cover over the feed and each of the three stages during the lab to prevent acetic acid from evaporating. Also, keep all reflux valves fully closed during the entire experiment. One last thing, the flowrates of the extract and raffinate can be measured using a graduated cylinder if needed.
3. Next, establish a decent initial level of water (i.e. solvent/extract) in all three stages, then adjust the water flowrate to 0.67 GPM on the water rotameter. Then, set all three agitators to their first rotation speed (three speeds will be used; **450 RPM, 650 RPM, and 850 RPM**).
    a. **NOTE:** The previous group had to use a water flowrate of ~0.80 GPM, we might have too as well.
4. Once water has been introduced into each of the three stages and has a steady flowrate; turn the centrifugal pump back on to begin feeding the white oil (i.e. mineral oil)/acetic acid solution to the unit. At this time, the valve to the feed rotameter should be **opened** and the valve that returns the feed solution back to the feed tank should be **closed**.

a. **NOTE:** As soon as the feed solution is allowed into the system; be sure to **time** how long it takes for the first amount of mineral oil (i.e. white oil) to reach the product tank. This will be the residence time for the oil system.
b. **CAUTION:** Do not run the centrifugal pump dry. Be sure the feed solution stays above the pump inlet.

5. Adjust the flowrate of the feed solution to ½ the flowrate of the water as this will allow for good mixing plus the development of an equilibrium interaction between the two phases and the solute.
6. Once adequate flowrates of the feed and water are attained, they should remain constant for the duration of the lab. Routine monitoring of the rotameters will be necessary.
    a. **NOTE:** There will be roughly 20 minutes available for each agitator speed to reach a steady state in each of the three stages. After steady state has been reached, samples need to be taken quickly. This calculation was done based on an expected max feed flowrate of 0.40 GPM and the fact that the feed tank will have ~25 gallons of solution.
7. For each trial, wait 12-16 minutes for the three stages to reach steady state, then take three 50 mL samples of both phases (water & white oil) for titration. Use parafilm to cover the samples while they wait to be titrated. Repeat this process for all three agitator speeds.
    a. **NOTE:** Due to water's higher density, it will form a phase on the bottom of each stage, while mineral oil (i.e. white oil) will form a phase above. Also, when titrating the mineral oil samples, use a stir plate and a stir bar to help mix the sample for better titration results.

Independent/Controlled Variables: agitator speed, feed composition, number of stages, extract/feed flowrate, solvent-to-feed flow ratio, no reflux, titrant (and its conc.), countercurrent operation, starting amount of feed, room temp/pressure, and tank/stage covers on.
Dependent Variables: acetic acid wt% in both phases for each stage, efficiency values, mass-transfer of acetic acid, and the white oil residence time.

POST-LAB PLAN:

1. Create plots of acetic acid concentration, for both phases (Y_A & X_A), as a function of stage number and agitator speed.
2. Plot Y_A (acetic acid in extract) vs. X_A (acetic acid in raffinate) for all three agitator speeds.
3. Perform a McCabe-Thiele analysis to acquire the number of theoretical stages needed for the liquid-liquid extractor. This will help determine the apparatus' efficiency.
4. Calculate both Murphree efficiencies for the three stages and the overall efficiency.

5. Explain the effect of the agitator speed on the efficiencies and acetic acid concentration.

Project Safety Evaluation (10pts)

Project Title:____LIQUID-LIQUID EXTRACTION___________________________

Date: _____4/7/2026____

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The agitator motors on each stage are electrically driven which present a shock risk if the liquid spills and interacts with the motor wiring.
Potential mechanical hazards?	Yes	If yes, identify conditions: The rotating agitator shafts present a pinch-point hazard. Loose clothing and hands should be kept away from shafts during operations.
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: At proper concentrations, acetic acid is corrosive and can cause severe skin burns, eye damage, and respiratory irritation. NaOH is caustic and highly dangerous. Mineral oil is mild irritant and carcinogenic.
Gloves required?	Yes	If yes, specify type: Nitrile gloves are required when handling NaOH, acetic acid, and other lab material and supply.
Other hazards	Yes	If yes, specify conditions: Slipping Hazards from the oil • Acetic acid vapor can irritate the respiratory tract. • Keep the stages and feed tank covers on except when obtaining samples. • Mineral oil spills can cause slipping hazards.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: • Sodium hydroxide (NaOH) — **CAS: 1310-73-2**

		• Phenolphthalein — **CAS: 77-09-8** • Acetic Acid **(**CH_3COOH**)** - **CAS: 64-19-7** • Mineral Oil (aka white oil) **– CAS: 8042-47-5**
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	25 °C	**Location:** All three stages and the general lab space **Hazards:** Little hazards since near room temperature **Precautions:** Standard PPE equipment must be used
Pressure (psia)	~14.7 psia (lab) ~30.0 psia (process)	**Location:** In the general lab space and within the process **Hazards:** Running the pump dry could damage it **Precautions:** Ensure the feed tank maintains the minimum level required for proper pump operation
Safety Controls	*yes/no*	*If yes, Provide description*
Rotameter Flow Controls	*Yes*	*Measures the water and mineral oil flowrates during the lab. They also allow for the manipulation of both flowrates to a desired reading.*
Reflux valves	*Yes*	*All the reflux valves must stay closed for all trials, ensuring consistent operation for all agitator speeds.*
Feed tank and stage covers	*Yes*	*These must stay on during operation to prevent acetic acid from evaporating.*

SAFETY STATEMENT:

I have read relevant background material for the Unit Operations Laboratory entitled: "LIQUID-LIQUID EXTRACTION" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance with standards and acceptable safety practices and will conduct all my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Team Member Signature: Cesearo Dilosa — Date: 7 April 2026

Team Member Signature: Ify Bidokwu — Date: 7 April 2026

Team Member Signature: Joshua Laforest — Date: 7 April 2026

Team Member Signature: Javier Silva — Date: 7 April 2026